

2 TWO APP STUDIO

Two App Studio: Turning online journals into biographies and digital clones with Google Cloud AI

Get started for free

GOOGLE CLOUD RESULTS

- ✓ Increased average engagement per user by 19% with [Vertex AI](#) and [Gemini](#)
- ✓ Improved conversion rates to paid subscriptions by 8% with Gemini and Vertex AI
- ✓ Increased development and operational teams' productivity by 30% with [Google Cloud and Firebase](#)
- ✓ Reduced operational costs by 20% with Google Cloud and Firebase
- ✓ Created opportunities to develop 'grief-tech' products such as digital clones with Gemini and Vertex AI

Two App Studio, an app development business in Singapore leverages Google Cloud, Firebase, Gemini, and Vertex AI to scale their mobile app, Journey, offer AI assistance, and explore new opportunities.

Keeping a journal can help individuals establish meaning and context to their lives, and provide a detailed reference for family and loved ones when they have passed away. Cloud computing has transformed this historic practice by giving mobile journal app developers access to scalable, reliable and easy to use infrastructure that supports voice, video and text. The rapid emergence of AI products and services such as [Vertex AI](#) and [Gemini](#) from [Google Cloud](#) adds a rich new dimension for journal app users to learn more about themselves by questioning chatbots. In addition, app developers can turn journals into biographies to be read and retained by future generations, and explore opportunities known as 'grief tech,' including digital clones of people who have passed away.

[Two App Studio](#), headquartered in Singapore, has turned to Google Cloud to add these new AI-powered features to its [Journey](#) mobile journal and diary app. The six-person business, established in 2016 to fill what its founders claim was a gap in the market for a private, accessible, and secure journal app, now provides Journey to millions of users across Android, Chrome, iOS, macOS, Linux, and the web.

The business's interest in generative AI sparked into life when Chief Executive Officer Jeromy Yap read online about a coder and data scientist's experience feeding her childhood diaries into a popular virtual assistant to train it to emulate her inner child. "I began to think about whether we could bring a capability like this into Journey, and asked our marketer to feed her own diaries into the same assistant and write up the experience," says Yap. "The insights the marketer derived prompted us to move forward in using Google Cloud AI to move our value proposition from capturing and saving memories to enabling users to learn more about themselves."

“The insights the marketer derived prompted us to move forward in using Google Cloud AI to move our value proposition from capturing and saving memories to enabling users to learn more about themselves.”

Jeromy Yap

Chief Executive Officer, Two App Studio



Gemini’s versatility and adaptability powering Odyssey AI

In late 2023, Two App Studio launched a digital assistant called Odyssey AI that relies on Gemini and Vertex AI to enable functions that add depth and relevance to the Journal app. These include optical character recognition for image categorization, the automated generation of image captions, and search using the function-calling feature of Gemini. The service also provides human-like responses to user queries to support natural interactions with journal entries.

“Gemini has shown great versatility and adaptability in powering innovative features that meet user needs, such as personalized content in Journey,” says Yap. “For instance, it customizes photo categories based on the top photo themes unique to each user. This means that each user sees photo categories that are most relevant to them, making their journal experience more personalized and engaging.”

The use of Vertex AI and the Gemini API for smart photo categories and Odyssey AI has delivered a considerable boost to user engagement for Two App Studio. “After a recent update to our Android app, we increased average user engagement per user by 19%, while the number of views per user went up by over 7%,” says Yap. “Conversion rates to paid subscriptions went up by 8%. We are

now advancing our plans to use generative AI to preserve and share users' life stories with the next generation—effectively turning journals into biographies that can be printed out and read by users' families.”

Two App Studio is also exploring options to turn individuals' journals into digital clones that enable their family to continue to engage with their thoughts, philosophy, and personality after they have passed away. Furthermore, the organization is evaluating opportunities to transform static memories into, for example, an animated, four-second video, and to color black and white journal photos. “We have become a lot more ambitious for our product when we found out what we could do with generative AI,” says Yap.

“We are now advancing our plans to use generative AI to preserve and share users' life stories with the next generation—effectively turning journals into biographies that can be printed out and read by users' families.”

Jeromy Yap

Chief Executive Officer, Two App Studio



Launching AI services from a reliable, low-cost infrastructure

Two App Studio's AI plans are being made considerably easier by the business's decision to run Journey on a Google Cloud and [Firebase](#) infrastructure. The organization had started operations on a traditional cloud service that made it difficult for its administrators to correctly provision virtual-machine instances to support demand. This forced Two App Studio to lean towards over-provisioning its instances and keep skilled team members on standby to resolve issues 24 hours a day, seven days a week, hampering its ability to control costs. The business also wanted to reduce the resources required to keep security patches up to date,

and move to a more closely integrated service to minimize its reliance on API keys.

After a comprehensive market review, Two App Studio identified Google Cloud and Firebase as the best fit for its needs, due to ease of development and maintenance, lower cost, and superior security. “The Firebase services combined seamlessly to reduce our need to share sensitive keys, minimizing the risk of breaches through human error,” says Yap.

Tight integration between the two platforms made development simple, with Yap citing the ease of setting up [Cloud Functions for Firebase](#) to trigger backend code in response to events, with [Cloud Tasks](#) to manage distributed tasks.

In addition, the ability to develop across multiple platforms, including Android, iOS, and the web, aligned with the organization’s direction of making its app widely accessible to users.

The functionality of individual services, including [Firebase Authentication](#) to manage login and [Firebase Local Emulator Suite](#) to enable testing of complex scenarios locally before going live, was also attractive to Two App Studio. Furthermore, Google Cloud offered comprehensive guides, sample code, and a strong community, simplifying the learning and implementation process for developers.

Using Google Cloud and Firebase also means resources scale seamlessly to align with demand, so Two App Studio pays only for what it uses and avoids upfront cost and waste. The business has

“Google Cloud is helping us realize our goal with its strong, scalable cloud services that keep our users’ data safe and accessible from any location. This allows our users to write their journals without worrying about privacy or losing their data.”

Jeromy Yap

Chief Executive Officer,
Two App Studio

set up alerts to avoid unexpected expenses and tracks spending using detailed billing reports. In addition, because Google Cloud automates updates, the business does not need to keep team members available 24 hours a day, seven days a week. Monitoring tools provide real-time insights into app performance to enable rapid identification and remediation of any issues, while automated backup and disaster-recovery options help minimize risk.

With Google Cloud and Firebase, Two App Studio's development and operational teams have increased productivity by 30%.

"Google Cloud has streamlined decision-making and created more efficient workflows," explains Yap. "Instead of managing multiple service accounts, we use [Identity and Access Management \(IAM\)](#) to assign and grant permissions to developers based on their current projects. This removes the need for management to create accounts or generate API keys for different services, letting us implement new features quickly."

The organization has reduced the time required for system upgrades and maintenance from an average of 12 manpower hours per week to less than eight. "Now, we need only focus on updating libraries and dependencies, saving a lot of time and effort," says Yap.

Automation through [Cloud Run functions](#) and [Cloud Scheduler](#) has reduced the time required for routine maintenance tasks by 40%, enabling the team to focus on more innovation and improving user experience. Adopting a pay-as-you go model and optimizing resource usage has reduced operational costs by 20%, with the savings invested in enhancing the app. The business is also yet to experience any downtime on Google Cloud and Firebase, ensuring a high-quality user experience and avoiding potential reputational damage and lost business arising from a significant outage.

"Google Cloud is helping us realize our goal with its strong, scalable cloud services that keep our users' data safe and accessible from any location," concludes Yap. "This allows our users to write their journals without worrying about privacy or losing their data. Google Cloud enables us to maintain our infrastructure easily and expand it quickly to keep up with user demand. Meanwhile, Gemini with Vertex AI has greatly improved how users experience and engage with our app—they can ask for information in their own way rather than rely exclusively on the app interface."

Headquartered in Singapore, **Two App Studio** develops popular in-house apps, including journal and diary app Journey that

empowers millions of users to record their thoughts and feelings.

Industry: Media & Entertainment

Location: Singapore

Products: [Gemini](#), [Vertex AI](#), [Cloud Functions for Firebase](#), [Cloud Run functions](#), [Cloud Scheduler](#), [Cloud Tasks](#), [Firebase Authentication](#), [Firebase Local Emulator Suite](#), [Identity and Access Management \(IAM\)](#).

Why Google	Products and	Solutions	Resources	Engage
Choosing Google Cloud	pricing	Infrastructure modernization	Google Cloud Affiliate Program	Contact sales
Trust and security	Google Cloud pricing	Databases	Google Cloud documentation	Find a Partner
Modern Infrastructure Cloud	Google Workspace pricing	Application modernization	Google Cloud quickstarts	Become a Partner
Multicloud	See all products	Smart analytics	Google Cloud Marketplace	Events
Global infrastructure		Artificial Intelligence		Podcasts
Customers and case studies		Security	Learn about cloud computing	Developer Center
Analyst reports		Productivity & work transformation	Support	Press Corner
Whitepapers		Industry solutions	Code samples	Google Cloud on YouTube
Blog		DevOps solutions	Cloud Architecture Center	Google Cloud Tech on YouTube
		Small business solutions	Training	Follow on X
		See all solutions	Certifications	Join User Research
			Google for Developers	We're hiring. Join Google Cloud!
			Google Cloud for Startups	Google Cloud Community
			System status	
			Release Notes	